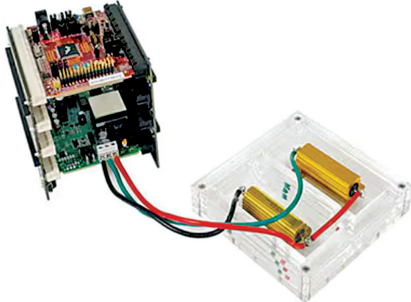
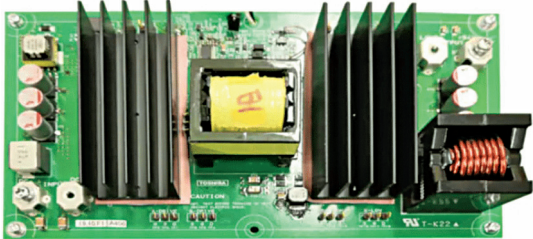


Interesting Reference Designs Of FULL-BRIDGE CONVERTER

Title of Reference Design	Full-Bridge DC-DC SMPS	Full-Bridge DC-DC Converter
Image		
Key Attractions	The reference design supports testing various power control methods. It provides real-time voltage and current feedback. A load circuit enables users to test performance under different load conditions. This helps develop and refine converter systems.	The 1kW full-bridge DC-DC converter reference design delivers efficient power with a stable output. It uses high-frequency switching, wide input range, and isolated communication, making it suitable for compact server power supply applications.
Highlights	NXP Semiconductors provides a full-bridge DC-DC switch-mode power supply (SMPS) reference design that shows power conversion using V series microcontrollers (MCUs). Operating as a transformer-isolated buck converter, the design comprises a full-bridge inverter, a transformer, a rectification stage, and an output filter. The low-voltage full-bridge DC-DC SMPS tower board works as a module in the tower system, allowing development and prototyping of power-control topologies. Developers can use two MCU/peripheral boards from the tower system to set control strategies such as peak current mode, average current mode, and voltage mode control. The design includes feedback circuits for operation, with voltage feedback provided through a resistor divider and current feedback achieved through a current transformer with a load resistor, both of which are adjusted for MCU sensing. A load circuit module is included to support testing and checking of power-control methods and software under different load conditions using a connected output resistor.	The 1kW full-bridge DC-DC converter reference design by Toshiba offers a power conversion solution for server power supplies, using a phase-shifted full-bridge topology with synchronous rectification to deliver output and efficiency in a compact form. It operates with an input voltage range of -36V to -60V DC and provides a regulated 54V DC output, delivering up to 1kW of power with a total efficiency of 94.7% at full load. Components include power MOSFETs, four N-channel MOSFETs on the primary side rated at 80V with low on-resistance using U-MOSX-H process, and eight N-channel MOSFETs on the secondary side rated at 150V with 0.0154Ω on-resistance at 10V using U-MOSVIII-H technology, enabling high-frequency switching and improving efficiency and thermal performance. The design also features two high-speed photocouplers for isolated communication between primary and secondary sides, supporting data rates up to 20Mbps.
Applications	The reference design is suitable for simple power applications.	The reference design is suitable for energy-conscious data centres and telecommunications applications.
OEM Brand	NXP Semiconductors	Toshiba
URL	https://www.electronicsforu.com/electronics-projects/full-bridge-dc-dc-smps-reference-design	https://www.electronicsforu.com/electronics-projects/reference-design-for-full-bridge-dc-dc-converter